A product obstruction to chiral quotients

Human Compression project

We use the previously established normalizer description of colour-preserving automorphisms of a connected coset flag graph. The following elementary consequence removes an entire class of regular covers before any subgroup enumeration. No priority claim is made for the observation.

Let a regular polytope have string C-group $W = \langle r_0, \dots, r_{d-1} \rangle$. Suppose a quotient polytope has flags Nw, with adjacency $Nw \mapsto Nwr_i$. The assumption that these are its actual flags is essential.

Lemma 1. *Suppose the colours partition into nonempty sets I, J, every generator in I commutes with every generator in J, and $W = W_I \times W_J$. Then*

$$N = (N \cap W_I) \times (N \cap W_J).$$

Proof. Write $n = ab \in N$ with $a \in W_I$ and $b \in W_J$. The flag $Na = Nb^{-1}$ is reachable from N both by an I-coloured path and by a J-coloured path. The first path preserves every face of rank in J; the second preserves every face of rank in I. Hence this flag contains precisely the faces of the flag N, so it is N itself. Thus $a, b \in N$. Every element of N therefore lies in the product of the two displayed intersections. The reverse inclusion is automatic, and the factors commute and meet trivially. □

Theorem 1. *A regular cover with an adjacent product of order two has no chiral polytope quotient with the stated coset flag action.*

Proof. Suppose $r_j r_{j+1}$ has order two. Since the generators are involutions, they commute. The string relations imply that every generator in $I = \{0, \dots, j\}$ commutes with every generator in $J = \{j+1, \dots, d-1\}$. The intersection property gives $W_I \cap W_J = 1$, and generation gives $W = W_I \times W_J$. By the lemma, $N = N_I \times N_J$, where $N_I = N \cap W_I$ and $N_J = N \cap W_J$. Consequently

$$N_W(N) = N_{W_I}(N_I) \times N_{W_J}(N_J).$$

If the quotient were chiral, its automorphism group would have exactly two flag orbits, and every adjacency would exchange them. Any product $r_i r_k$ of two generators therefore returns the base flag to its automorphism orbit, so $r_i r_k \in N_W(N)$ by the normalizer theorem. Take $i \in I$ and $k \in J$. The displayed product decomposition of the normalizer implies $r_i \in N_{W_I}(N_I)$ and $r_k \in N_{W_J}(N_J)$. Varying i and k shows that all generators normalize N. Thus $N_W(N) = W$, and the quotient has one flag orbit, a contradiction. □

The same proof applies to any partition satisfying the lemma, whether or not it arises from a single break in a string. It does not assert that an arbitrary subgroup of a direct product splits; the flag interpretation is what forces that conclusion. Nor does it identify an arbitrary orbit poset with its coset graph.

For ranks one and two there is no separate chiral case. A connected rank-two flag graph is an alternating cycle; rotations by two edges and a reflection preserving colours act transitively on its vertices. Rank one has a single edge. Thus the remaining small-cover search may restrict to rank at least three and to Schlaefli entries all at least three, while still checking the polytopal quotient condition.

This argument uses the source's regular-cover framework and the previous normalizer proof. It is a general exclusion, not a completed atlas census or a solution of a new open question. Completeness of the remaining finite cover roster remains an explicit input obligation.